

# Rafferty Simms

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## Work Experience

- **Platform Engineer - HPC & Linux** — Red Bull Racing & Red Bull Technology [May 2026 – Present]
  - Building and maintaining scalable platform infrastructure and services in a hybrid cloud environment to support F1 engineering workloads.
- **Information Technology Graduate** — Mercedes AMG High Performance Powertrains [Sep 2025 – May 2026]
  - Developed automation and software solutions (.NET, Python, SQL) supporting F1 powertrain R&D and business systems, saving ~10 weeks annually in engine build planning.
  - Maintained CI/CD pipelines and automation infrastructure using Azure DevOps and Ansible.
  - Administered servers, storage, networks, and Active Directory (provisioning, patching, IAM).
  - Conducted data risk assessments, producing threat analyses and risk acceptance documentation.
- **Information Technology Placement** — Mercedes AMG High Performance Powertrains [Aug 2023 – Sep 2024]
  - Led automation projects for license usage optimisation and power consumption reduction.
  - Supported F1 engineering infrastructure, including HPC analysis, maintenance, and operational continuity.
  - Maintained company-wide infrastructure reliability and security, including vulnerability remediation.
  - Built internal tools to monitor resources, ensure data integrity, and streamline workflows.

## Education & Certifications

- **Bachelor of Science, BSc (Hons) – Computer Systems (Cyber Security)** [Sep 2021 – May 2025]  
Nottingham Trent University – Grade: **First Class Honours** (88% average)
  - *Elective Course: University Language Programme, Chinese (Mandarin)* — Grade: *Distinction*
- **Certified Kubernetes Administrator (CKA)** – CNCF Certification [Jul 2026]
- **Certified Kubernetes Application Developer (CKAD)** – CNCF Certification [Aug 2026]
- **Certified in Cyber Security (CC)** – ISC2 Certification [Jul 2023]
- **Security, Compliance, and Identity Fundamentals** – Microsoft Certification [Jun 2023]

## Personal Project Highlights

- **Gait Recognition-based Authentication for Physical Access Control:** A machine learning-driven computer vision framework for gait recognition, integrated into a distributed physical access control system. Uses deep learning to analyse walking patterns and unobtrusively authenticate individuals during their approach to an entrance, achieving 72% overall classification accuracy, with 99% precision on high-confidence (>75%) predictions in controlled tests. Includes a front-end web interface enabling biometric identity registration and access management.
- **Turntable Track Recognition** : A Raspberry Pi-based system built with Python and deployed in a dockerised microservice architecture, enabling integration of an analogue turntable into a smart-home environment. Music is streamed to a HTTP endpoint for Sonos integration, and music recognition is performed on 15-second segments to live-update stream metadata with the current track and log listening history to Last.fm.

## Technical Skills

- **Programming:** Python, C#/.NET, JavaScript/TypeScript, Go, Java, SQL, HTML/CSS.
- **Infrastructure & DevOps:** Kubernetes, Docker, OCI/Azure/AWS, Linux (RHEL/Ubuntu), Windows, VMware/OLVM, Bare Metal, Ceph/NetApp/IBM Storage, Ansible, Terraform, CI/CD.

## Awards

- **British Computer Society and Department of Computer Science Best in Year Prize** [July 2025]  
Awarded for Best Performance in Final Year of BSc (Hons) Computer Systems (Cyber Security)
- **Best Final Year Project Showcase Prize** [May 2025]  
Awarded by Industry Representatives for Best Final Year Project at the End-of-Year Degree Showcase
- **Department of Computer Science Prize for Outstanding Performance** [Sep 2023]  
Awarded for Best Performance in Year Two of BSc (Hons) Computer Systems (Cyber Security)
- **Ryland Consulting and Department of Computer Science Prize for Exceptional Performance** [Aug 2022]  
Awarded for Best Performance in Year One of BSc (Hons) Computer Systems (Cyber Security)